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Recommendation systems

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AI and IT in Industry 2025

Motivation

Person	Air fryer	Backpack	Broom	Car	Chair	Dishwasher	Frying pan	Gaming console	Hiking shoes	Lightbulb	Motorcycle	SUV	Scooter	Swimsuit	Umbrella	Waffle iron
Alice	–	–	✅	❌	✅	–	✅	–	–	✅	❌	❌	❌	–	–	–
Bob	❌	✅	–	–	–	❌	–	❌	✅	–	–	–	–	✅	✅	❌
Charlie	❌	✅	–	–	–	❌	–	❌	✅	–	–	–	–	✅	✅	?
David	–	–	✅	❌	✅	–	✅	–	–	✅	❌	❌	❌	–	–	–
Ellen	–	–	❌	✅	❌	–	?	–	–	❌	✅	✅	✅	–	–	–
Fabian	✅	❌	–	–	–	✅	–	✅	❌	–	–	–	–	❌	❌	✅
Gabriela	–	–	✅	❌	✅	–	✅	–	–	✅	❌	?	❌	–	–	–
Hillary	❌	✅	–	–	–	❌	–	❌	✅	–	–	–	–	✅	✅	❌
Ian	?	❌	–	–	–	✅	–	✅	❌	–	–	–	–	❌	❌	✅
John	–	–	❌	✅	❌	–	❌	–	–	❌	✅	✅	✅	–	–	–
Kim	–	–	❌	✅	❌	–	❌	–	–	❌	✅	✅	✅	–	–	–
Michael	❌	✅	–	–	–	❌	–	❌	✅	–	–	–	–	✅	✅	❌
Nancy	–	–	✅	❌	✅	–	✅	–	–	?	❌	❌	❌	–	–	?
Omar	❌	✅	–	–	–	❌	–	❌	✅	–	–	–	–	✅	✅	❌
Peter	✅	❌	–	–	–	✅	–	✅	❌	–	–	–	–	❌	❌	✅
Raphael	–	–	✅	❌	✅	–	✅	–	?	✅	❌	❌	❌	–	–	–
Susan	–	–	❌	✅	❌	–	❌	–	–	❌	✅	✅	?	–	–	–
Thomas	✅	❌	–	?	?	✅	–	✅	❌	–	–	–	–	❌	❌	✅
Ulrich	✅	❌	–	–	–	✅	–	✅	❌	–	–	–	–	❌	❌	✅
Valery	✅	❌	–	–	–	✅	–	✅	❌	–	–	–	–	❌	❌	✅
Will	❌	✅	–	–	–	❌	–	❌	✅	–	–	–	–	✅	✅	❌

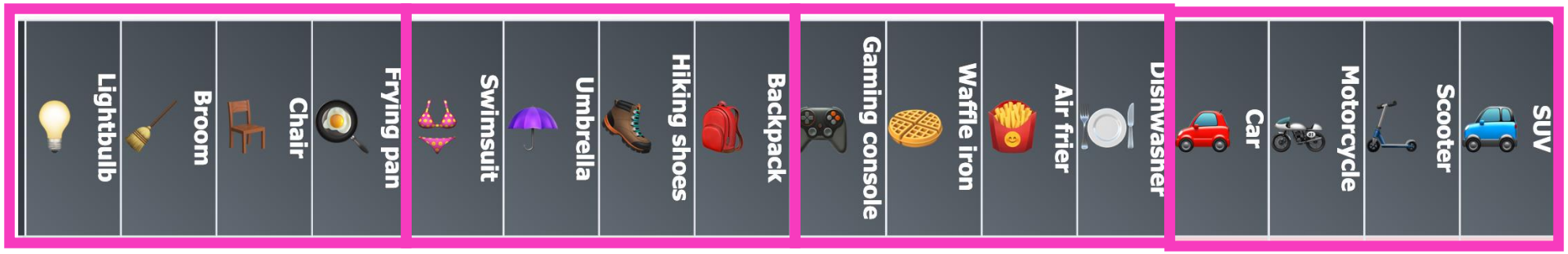
By rearranging items and people, we observe clusters

Person	Lightbulb	Broom	Chair	Frying pan	Swimsuit	Umbrella	Hiking shoes	Backpack	Gaming console	Waffle iron	Air fryer	Dishwasher	Car	Motorcycle	Scooter	SUV
Raphael	✅	✅	✅	✅	–	–	?	–	–	–	–	–	❌	❌	❌	❌
Nancy	?	✅	✅	✅	–	–	–	–	–	?	–	–	❌	❌	❌	❌
Alice	✅	✅	✅	✅	–	–	–	–	–	–	–	–	❌	❌	❌	❌
David	✅	✅	✅	✅	–	–	–	–	–	–	–	–	❌	❌	❌	❌
Gabriela	✅	✅	✅	✅	–	–	–	–	–	–	–	–	❌	❌	❌	?
Michael	–	–	–	–	✅	✅	✅	✅	❌	❌	❌	❌	–	–	–	–
Will	–	–	–	–	✅	✅	✅	✅	❌	❌	❌	❌	–	–	–	–
Bob	–	–	–	–	✅	✅	✅	✅	❌	❌	❌	❌	–	–	–	–
Omar	–	–	–	–	✅	✅	✅	✅	❌	❌	❌	❌	–	–	–	–
Hillary	–	–	–	–	✅	✅	✅	✅	❌	❌	❌	❌	–	–	–	–
Charlie	–	–	–	–	✅	✅	✅	✅	❌	?	❌	❌	–	–	–	–
Fabian	–	–	–	–	❌	❌	❌	❌	✅	✅	✅	✅	–	–	–	–
Ulrich	–	–	–	–	❌	❌	❌	❌	✅	✅	✅	✅	–	–	–	–
Thomas	–	–	?	–	❌	❌	❌	❌	✅	✅	✅	✅	?	–	–	–
Peter	–	–	–	–	❌	❌	❌	❌	✅	✅	✅	✅	–	–	–	–
Ian	–	–	–	–	❌	❌	❌	❌	✅	✅	?	✅	–	–	–	–
Valery	–	–	–	–	❌	❌	❌	❌	✅	✅	✅	✅	–	–	–	–
Susan	❌	❌	❌	❌	–	–	–	–	–	–	–	–	✅	✅	?	✅
Ellen	❌	❌	❌	?	–	–	–	–	–	–	–	–	✅	✅	✅	✅
John	❌	❌	❌	❌	–	–	–	–	–	–	–	–	✅	✅	✅	✅
Kim	❌	❌	❌	❌	–	–	–	–	–	–	–	–	✅	✅	✅	✅

A closer look at the items



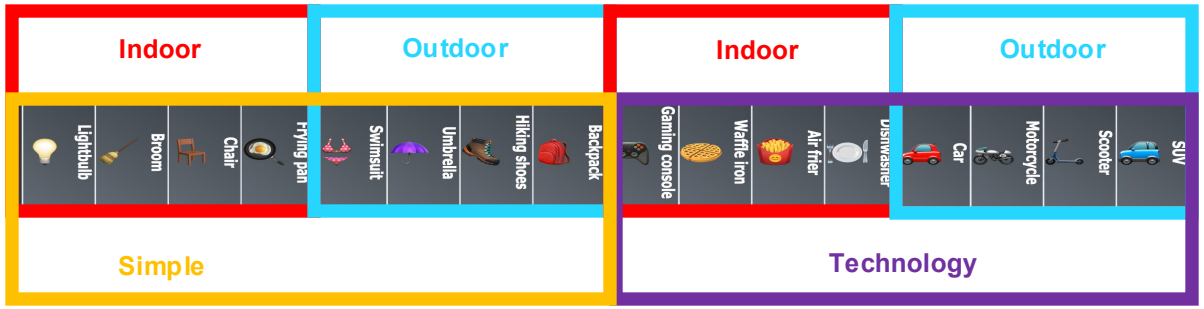
A closer look at the items



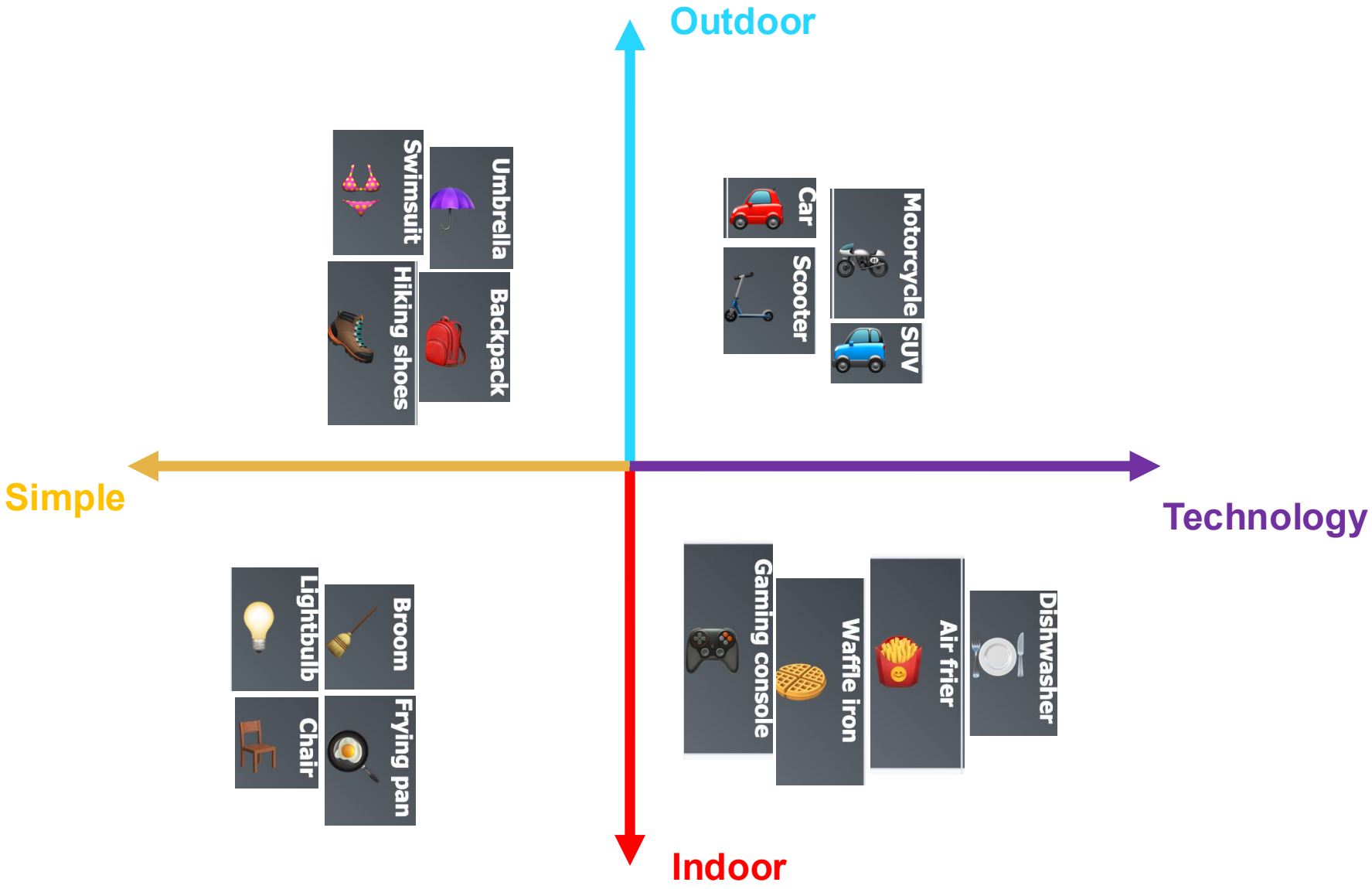
A closer look at the items



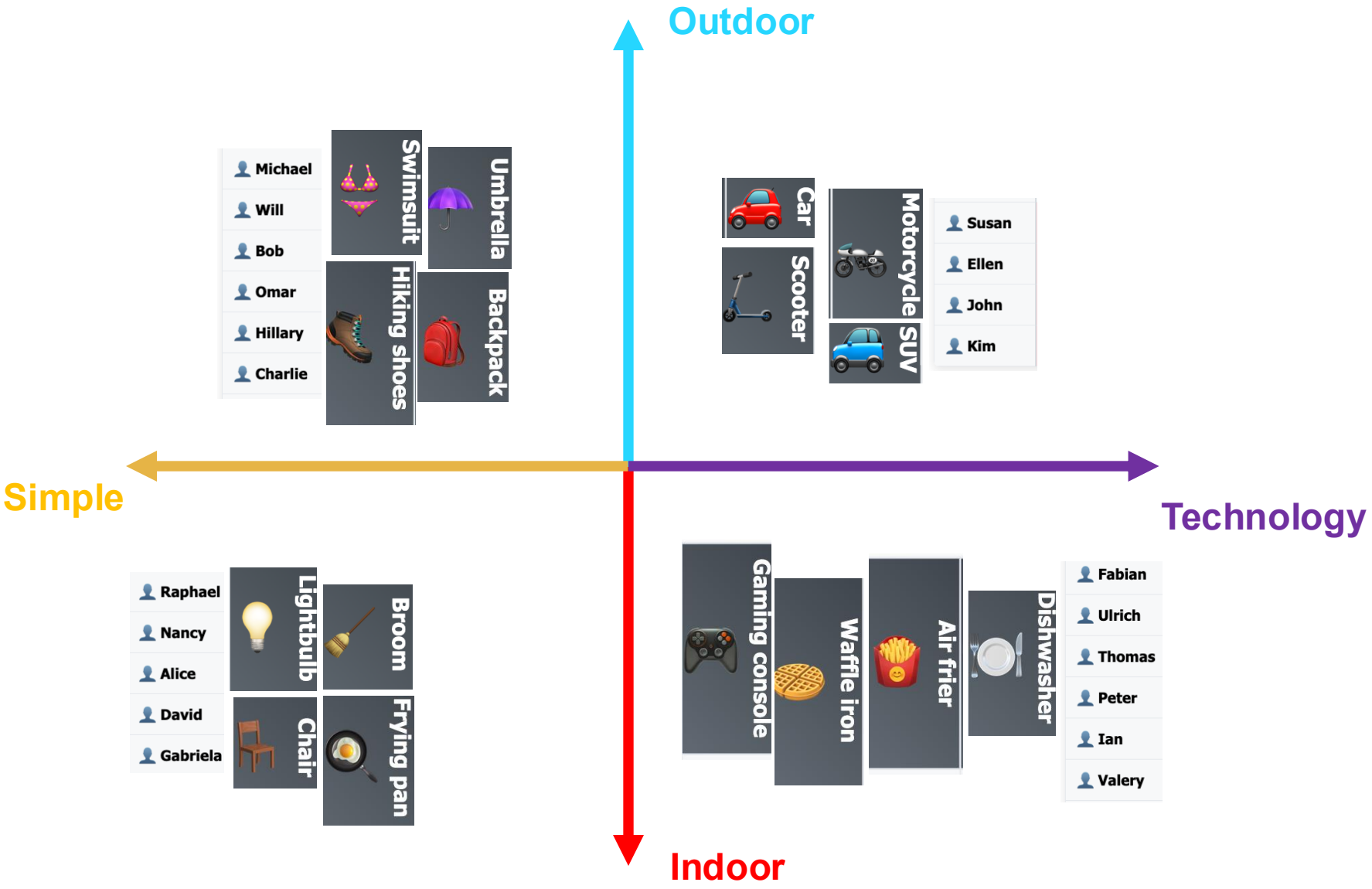
A closer look at the items



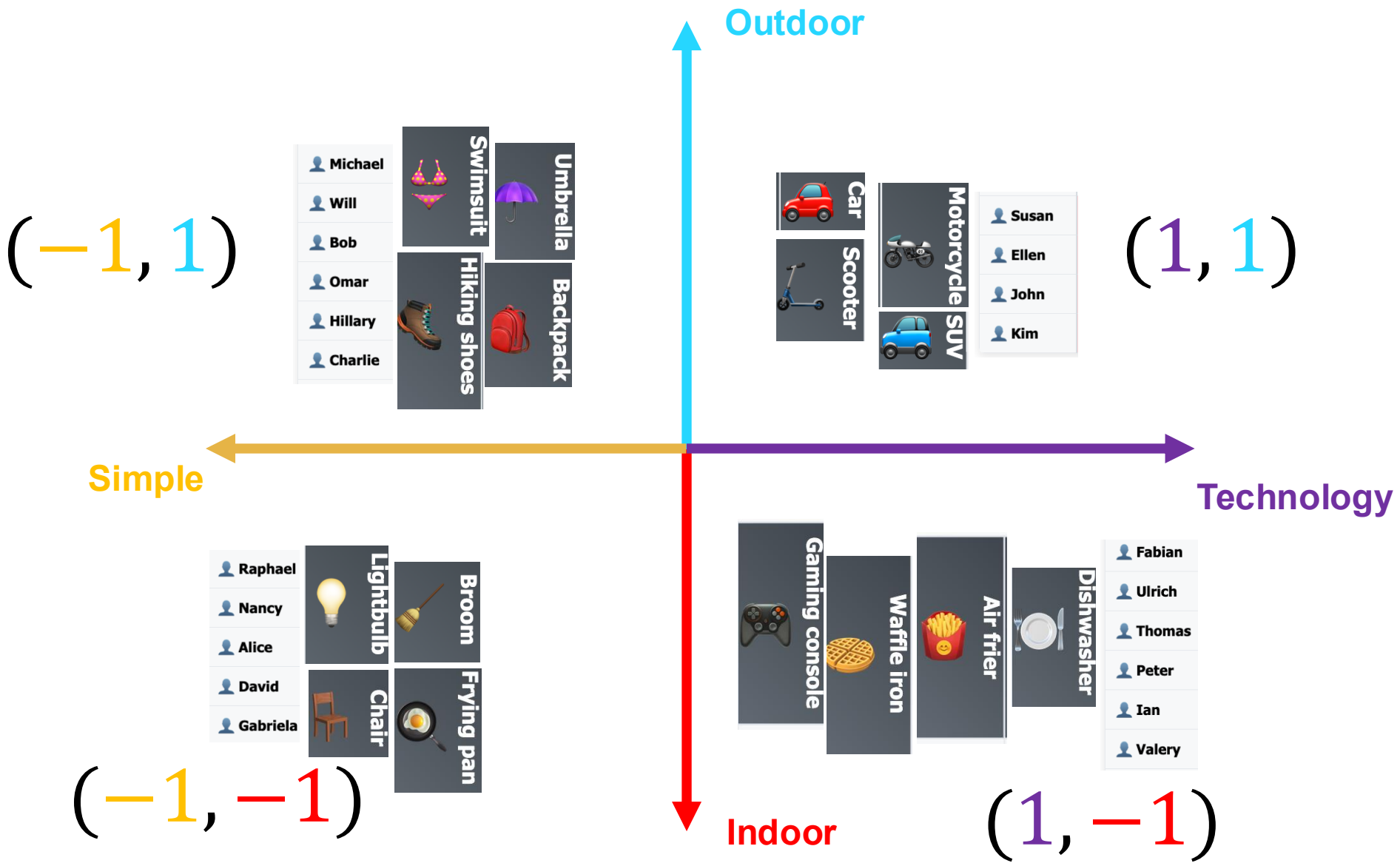
The items can be organized in an Euclidean space



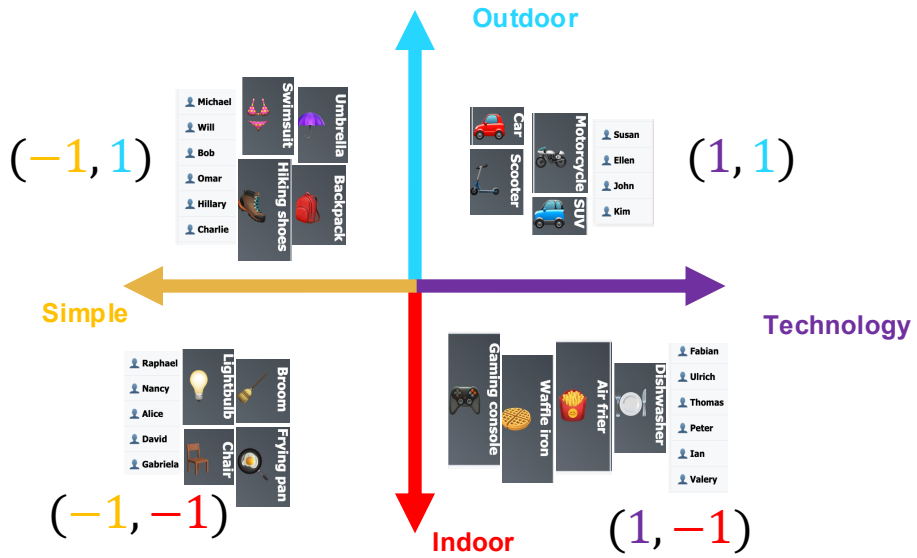
So the users can be matched to these items in this space



Users and items can be encoded as pairs



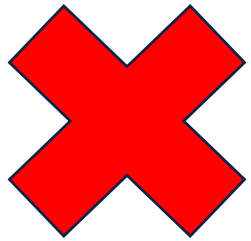
How much does Michael like the dishwasher?





$(-1, 1)$

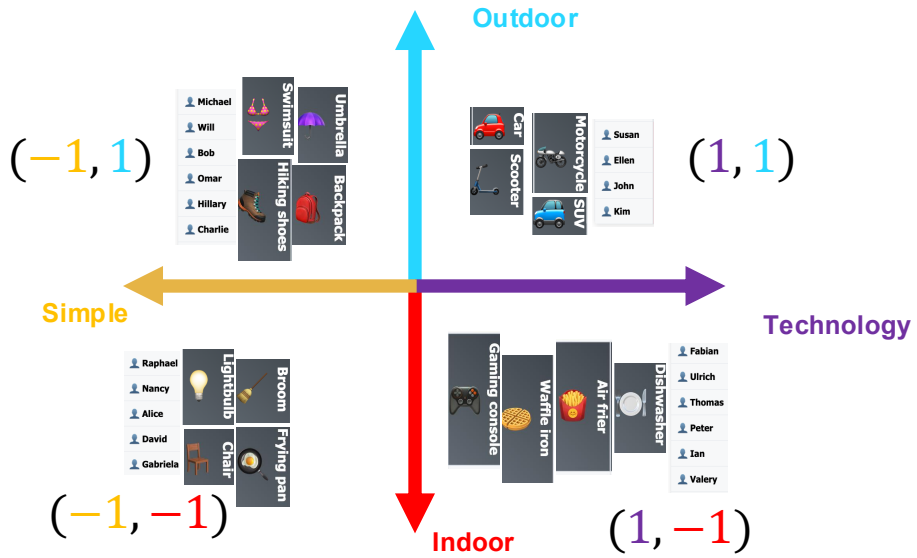
$$-1 \times 1 + 1 \times -1 = -2$$



$(1, -1)$



How much does Michael like the backpack?



Michael

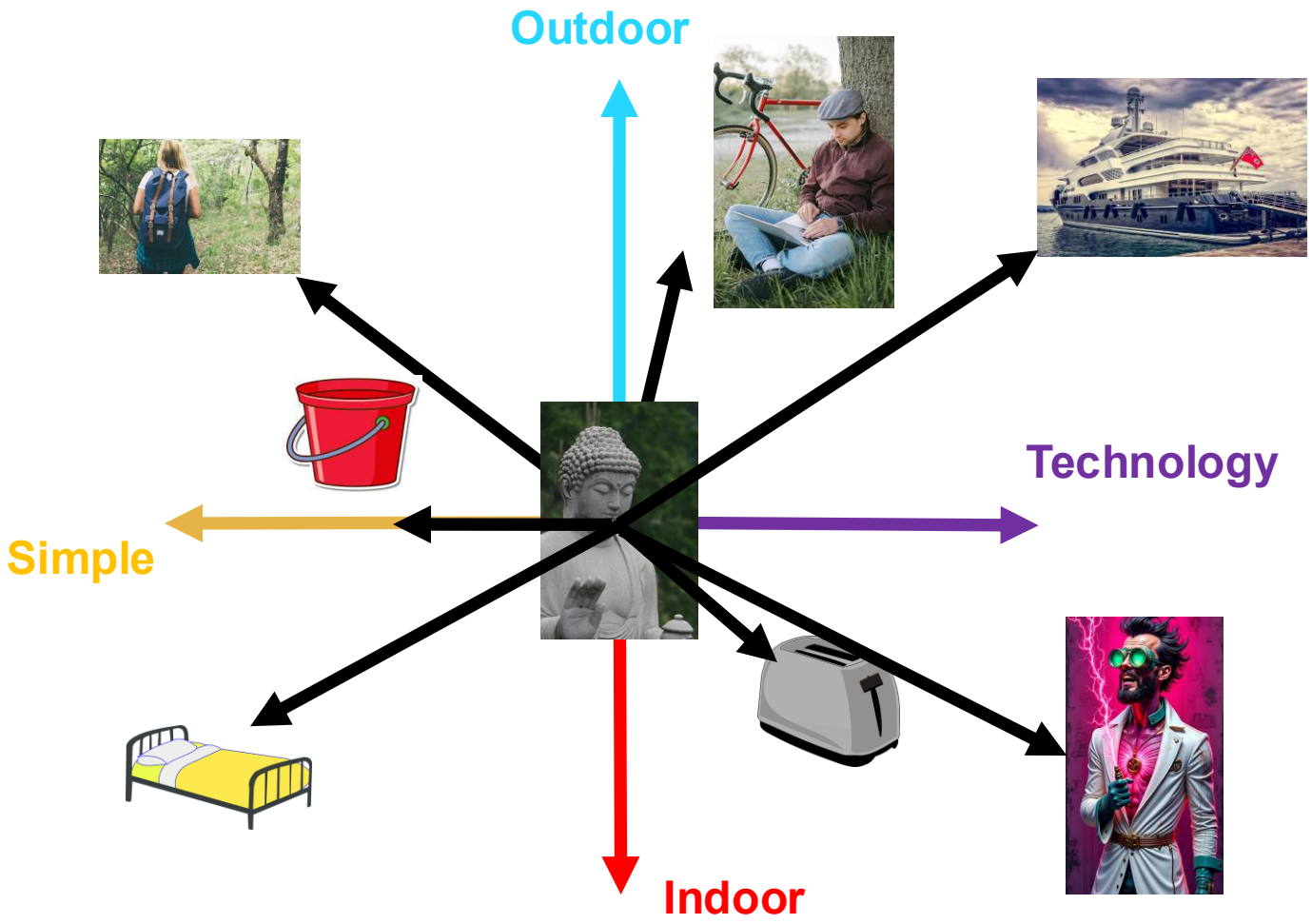
$(-1, 1)$

$$-1 \times 1 + -1 \times 1 = 2$$

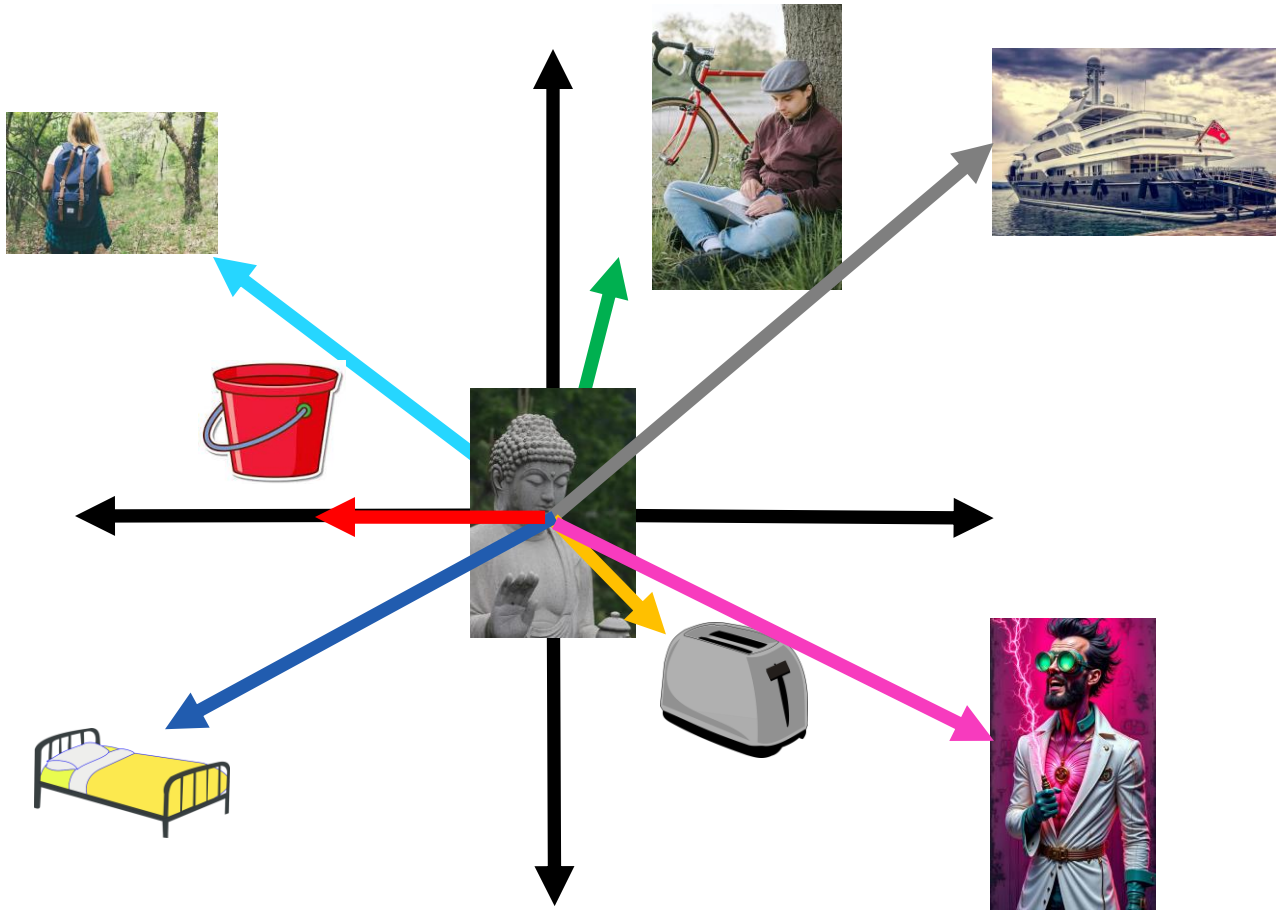


$(-1, 1)$

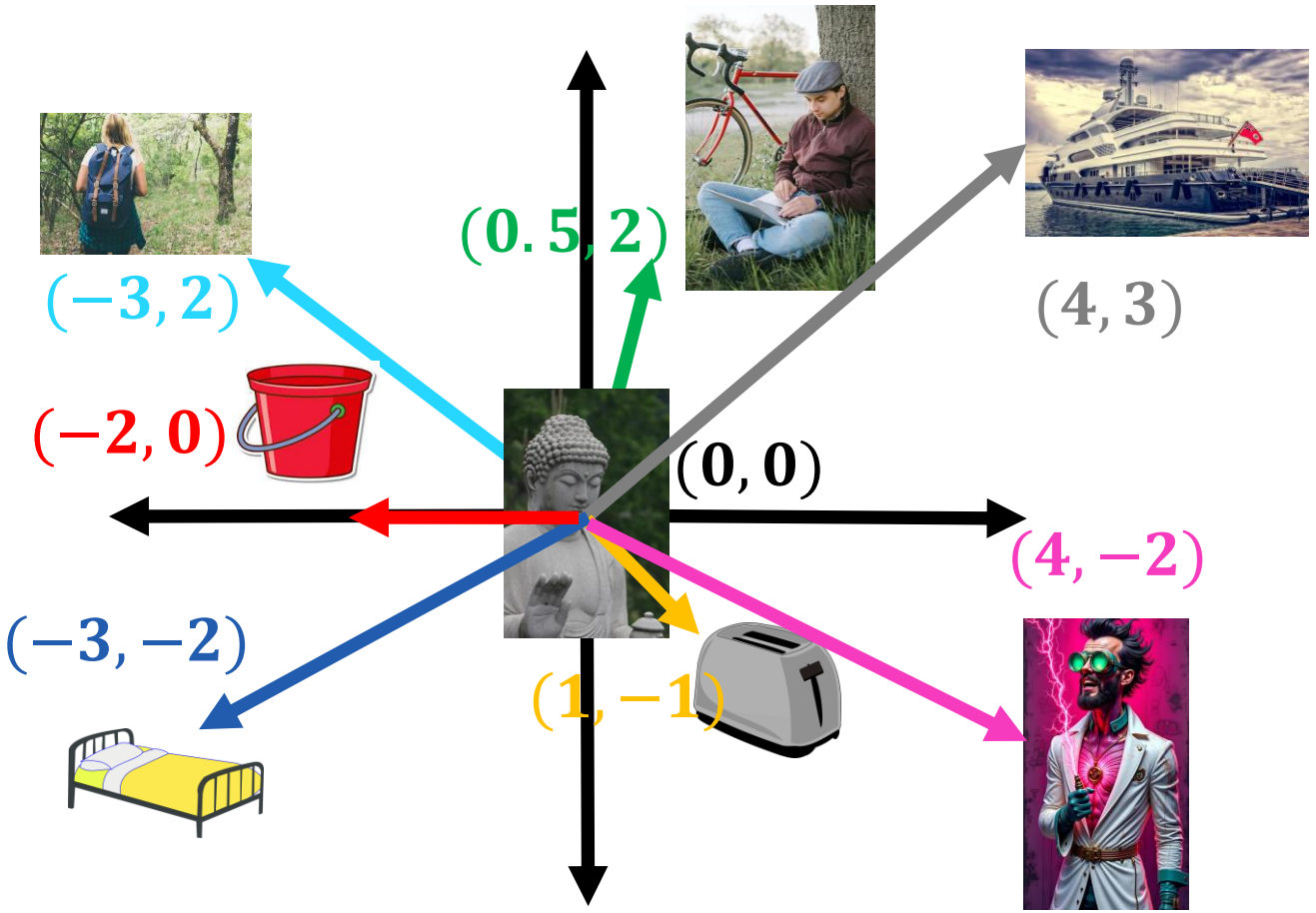
Users and items can be **embedded** into Euclidean spaces



Users and items can be **embedded** into Euclidean spaces



Users and items can be **embedded** into Euclidean spaces



The inner product

- For two vectors $x = (x_1, x_2)$ and $y = (y_1, y_2)$ in $\mathbb{R}^2$, we define their inner product as

$$x^\top y = x_1 y_1 + x_2 y_2.$$

The inner product

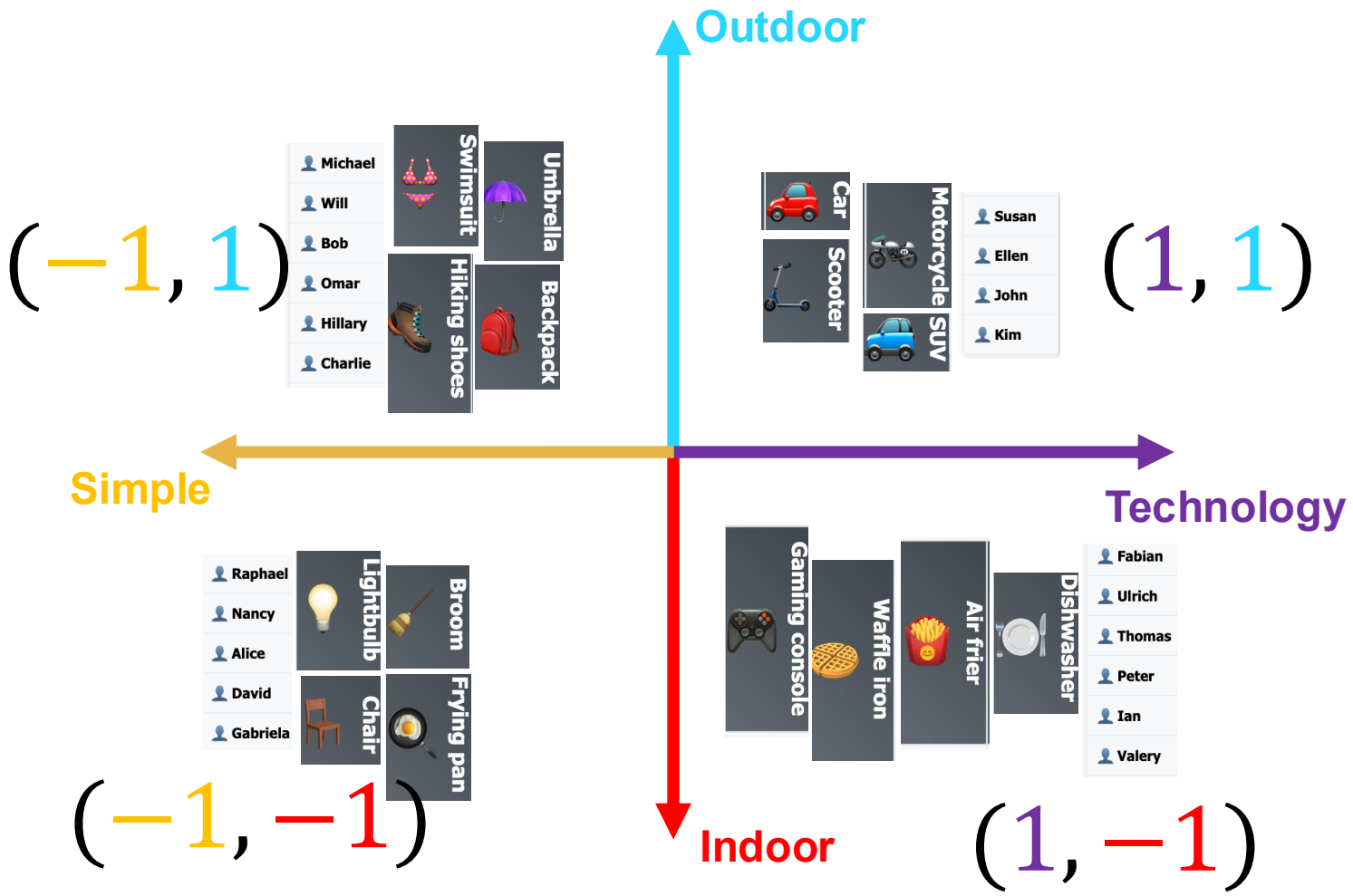
- For two vectors $x = (x_1, x_2)$ and $y = (y_1, y_2)$ in $\mathbb{R}^2$, we define their inner product as

$$x^\top y = x_1 y_1 + x_2 y_2.$$

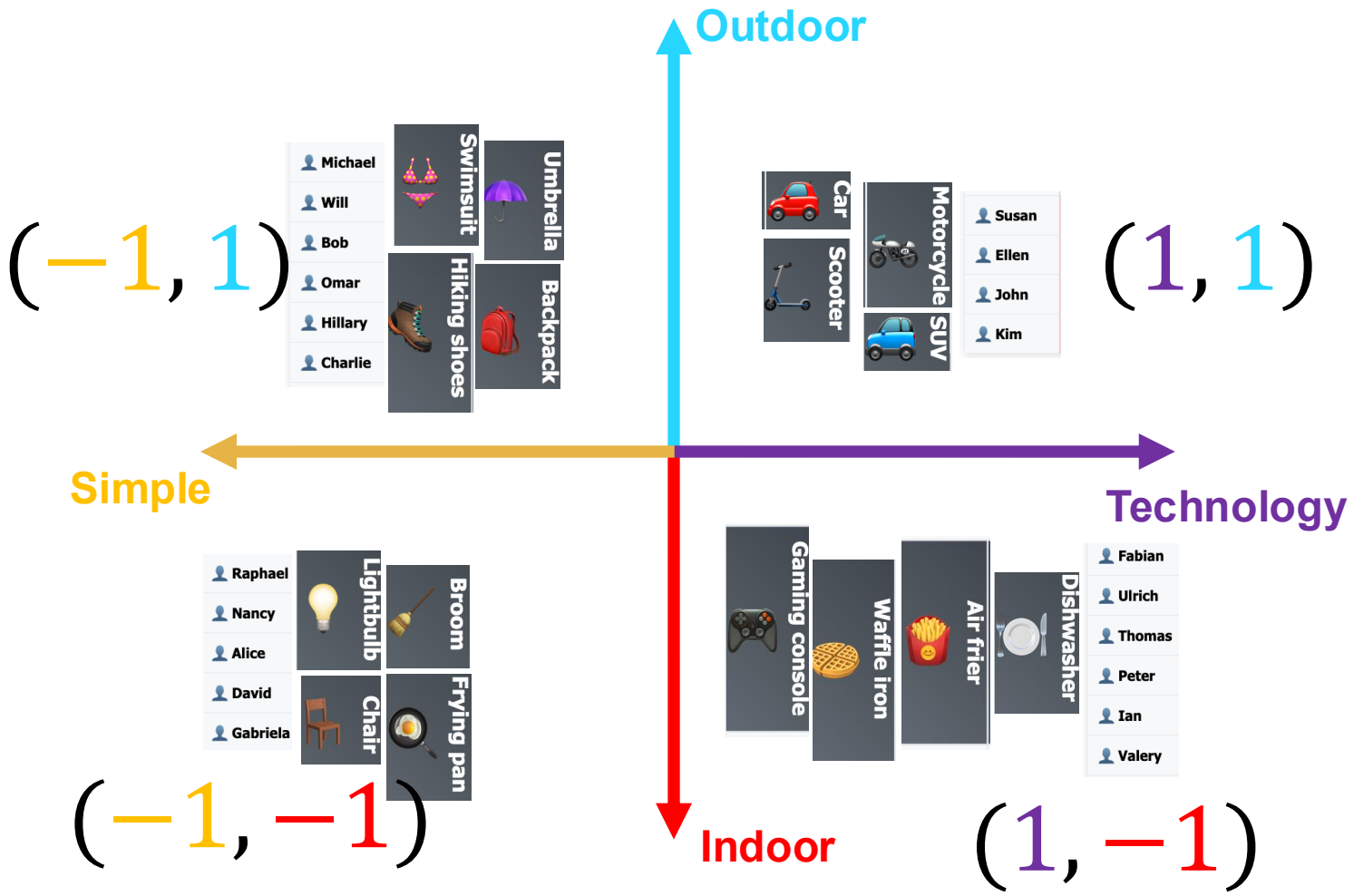
- More generally, for $x = (x_1, x_2, \ldots, x_d)$ and $y = (y_1, y_2, \ldots, y_d)$ in $\mathbb{R}^d$, we define their inner product as

$$x^\top y = x_1 y_1 + x_2 y_2 + \cdots + x_d y_d.$$

Measuring similarities with inner products



Measuring similarities with inner products

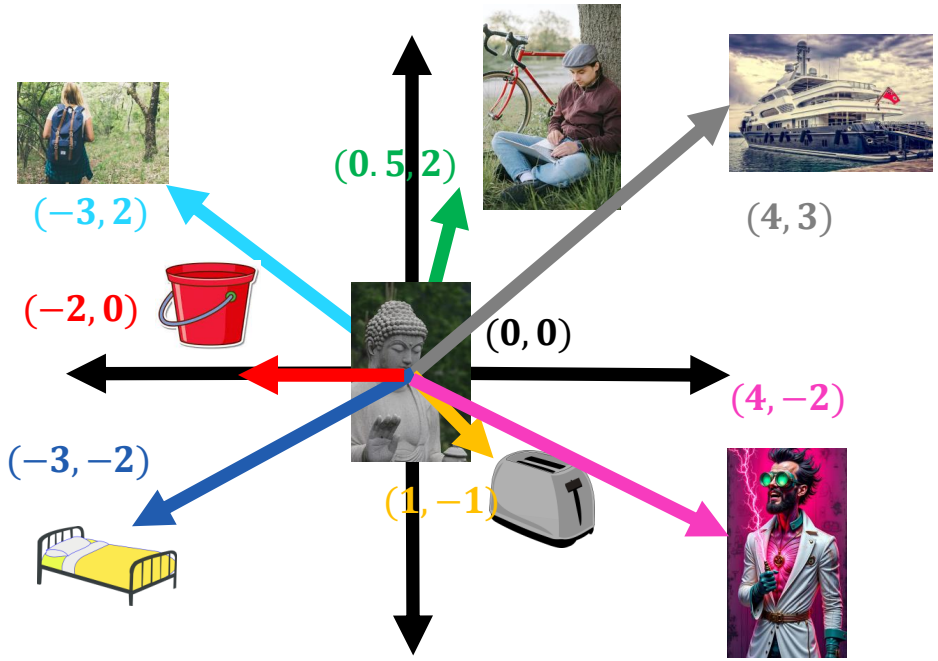


Person	Backpack	Gaming console	Waffle iron	Air frier	Dishwasher	Car	Motorcycle	Scooter	SUV
Raphael	–	–	–	–	–	✗	✗	✗	✗
Nancy	–	–	?	–	–	✗	✗	✗	✗
Alice	–	–	–	–	–	✗	✗	✗	✗
David	–	–	–	–	–	✗	✗	✗	✗
Gabriela	–	–	–	–	–	✗	✗	✗	?

The users as matrices and the items as matrices

		-1	-1	-1	-1	-1	-1	-1	-1	1	1	1	1	1	1	1	1
		-1	-1	-1	-1	1	1	1	1	-1	-1	-1	-1	1	1	1	1
-1	-1	Raphael	✅	✅	✅	✅	–	–	?	–	–	–	–	–	❌	❌	❌	❌
-1	-1	Nancy	?	✅	✅	✅	–	–	–	–	–	?	–	–	❌	❌	❌	❌
-1	-1	Alice	✅	✅	✅	✅	–	–	–	–	–	–	–	–	❌	❌	❌	❌
-1	-1	David	✅	✅	✅	✅	–	–	–	–	–	–	–	–	❌	❌	❌	❌
-1	-1	Gabriela	✅	✅	✅	✅	–	–	–	–	–	–	–	–	❌	❌	❌	?
-1	1	Michael	–	–	–	–	✅	✅	✅	✅	❌	❌	❌	❌	–	–	–	–
-1	1	Will	–	–	–	–	✅	✅	✅	✅	❌	❌	❌	❌	–	–	–	–
-1	1	Bob	–	–	–	–	✅	✅	✅	✅	❌	❌	❌	❌	–	–	–	–
		Person	Lightbulb	Broom	Chair	Frying pan	Swimsuit	Umbrella	Hiking shoes	Backpack	Gaming console	Waffle iron	Air frier	Dishwasher	Car	Motorcycle	Scooter	SUV

How matrices of preferences rise



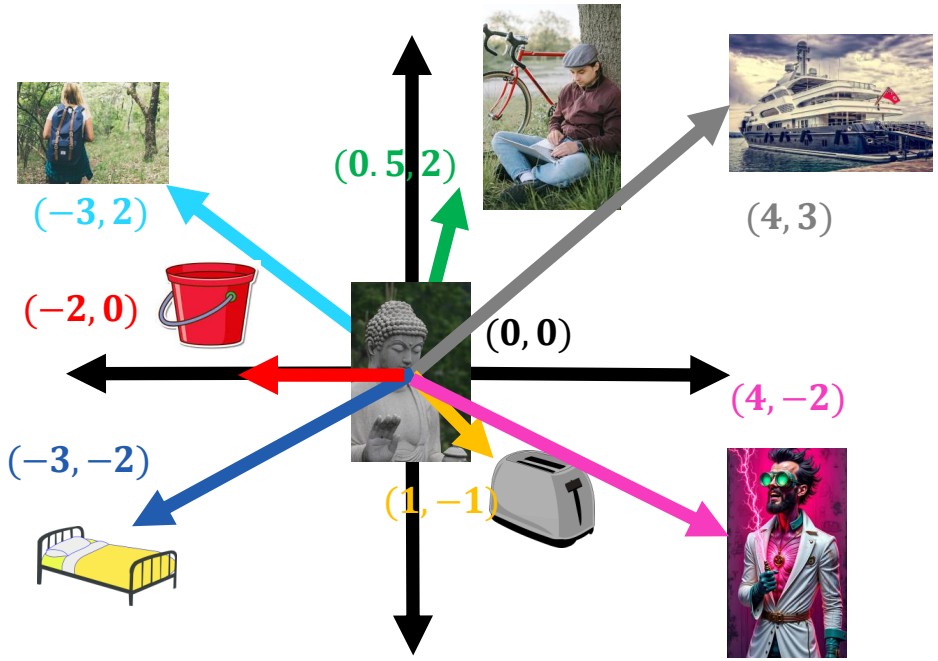


-2	4	1	-3
0	3	-1	2



-3	2
0.5	2
0	0
4	-2

How matrices of preferences rise








-3	2		
0.5	2		
0	0		
4	-2		

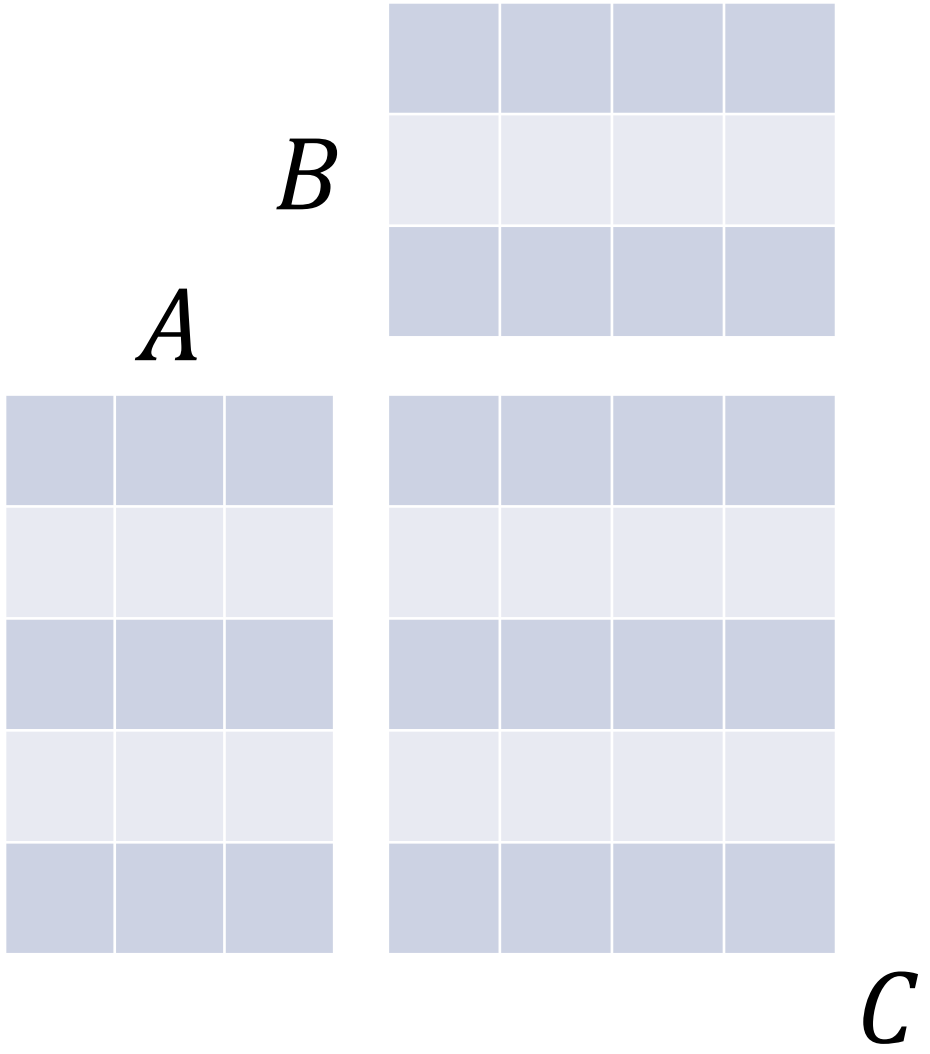





-2	4	1	-3
0	3	-1	2

6	-6	-5	13
-1	8	-1.5	2.5
0	0	0	0
-8	10	6	-16

Matrix multiplication



For two matrices A and B, their product $A \times B$ is a matrix C where

C_{ij} = inner product of row i of A and column j of B.

Matrix multiplication

B

1	-1	0	0
2	0	-2	0
0	-1	-1	1

A

1	2	0
-1	0	2
-2	-1	0
0	1	-2
1	0	-1

C

For two matrices A and B, their product $A \times B$ is a matrix C where

C_{ij} = inner product of row i of A and column j of B.

Matrix multiplication

B

1	-1	0	0
2	0	-2	0
0	-1	-1	1

A

1	2	0
-1	0	2
-2	-1	0
0	1	-2
1	0	-1

C

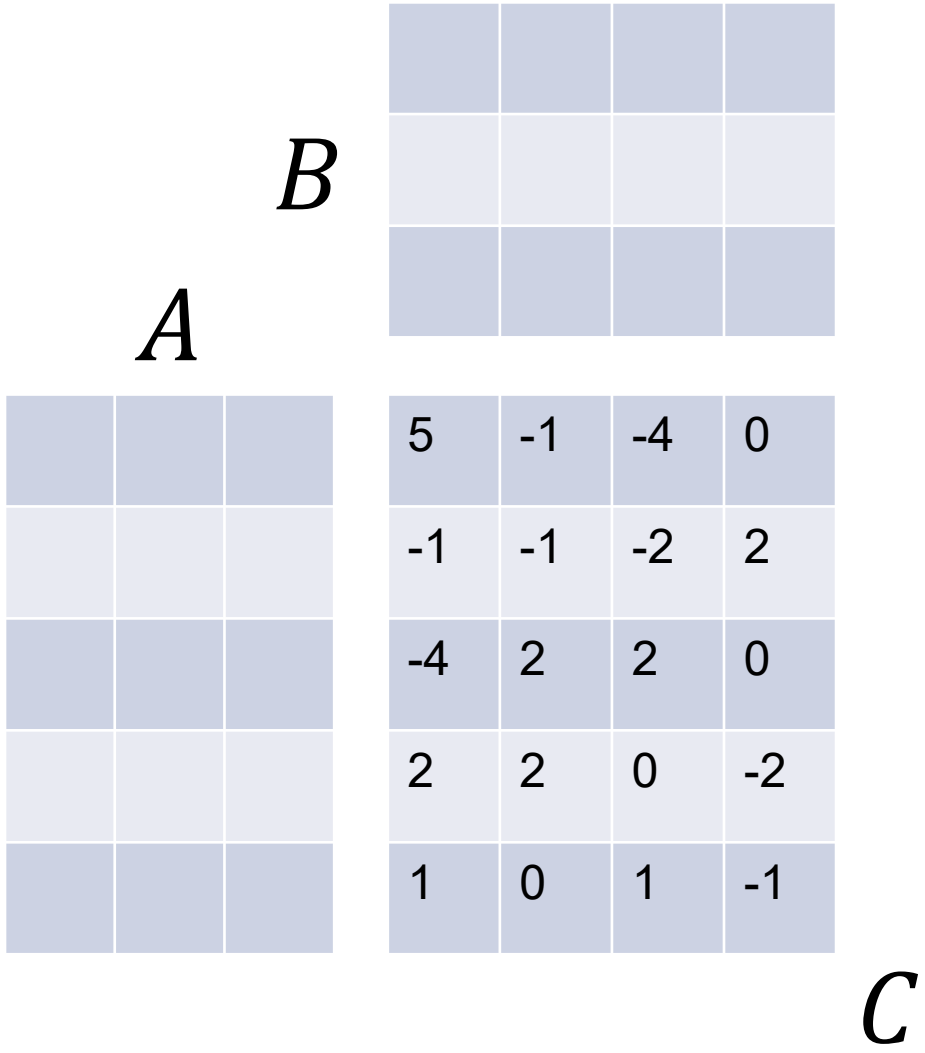
5	-1	-4	0
-1	-1	-2	2
-4	2	2	0
2	2	0	-2
1	0	1	-1

For two matrices A and B, their product $A \times B$ is a matrix C where

C_{ij} = inner product of row i of A and column j of B.

But wait a second… I only have the actual matrix of ratings, how do I get the user and item embedding matrices?

The matrix factorization problem



Given a matrix C, find two matrices A and B such that C = A x B.

Recommendation system as a matrix factorization problem

- We are given a matrix C with customer ratings of items.
- We must then find a factorization such that C = A x B.